Laboracle — Tech Stack, Tools & Pipelines

Full architecture inventory from day one to today: every framework, library, service, table, pipeline, and feature, with the user-facing benefit each one delivers.

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TOP-LEVEL APPS	MYSQL TABLES	LLM MODELS	AI MODES	OAUTH PROVIDERS	SUBSCRIPTION TIERS

Project journey, in order

How the codebase grew from a single FastAPI prototype into the current multi-provider, multi-mode, subscription-gated platform.

Foundation — Q&A on documents

Phase 1

FastAPI backend, React + Vite frontend, single OpenAI model (GPT-4o), FAISS vector store, RAG pipeline with `text-embedding-3-small`.

Outcome: students upload PDFs/DOCX/TXT and get cited answers.

Accounts, sessions, deployment

Phase 2

MySQL 8 user accounts, bcrypt passwords, bearer-token sessions, Docker Compose + AWS EC2 + nginx, phpMyAdmin for ops.

Outcome: real users, persistent chat history, production deploy.

Payments & tiers

Phase 3

Stripe Checkout + Customer Portal + webhooks, Free / Regular / Advanced tiers, monthly quotas for chat, uploads, and voice.

Outcome: monetisation, usage governance, upgrade path.

Auth providers & safety

Phase 4

Google / Facebook / Microsoft Sign-In, Cloudflare Turnstile CAPTCHA, slowapi rate limiting, password reset emails, IDOR hardening, file-size and MIME guards.

Outcome: lower signup friction, abuse protection.

Admin dashboard

Phase 5

Card-based dashboard with KPI strip, charts (signups, MRR, tier mix), user table with search & ban, AdGuard-proof CSS class names (`.adm-*`).

Outcome: ops visibility without a third-party BI tool.

Visual identity

Phase 6

Sticky frosted-glass navbar, hero with cyberpunk companion video on solid `#4E3D8C`, animated counters, conveyor-belt "How It Works" cards, bento-grid features, mouse-fleeing dot motif on auth.

Outcome: a marketing front page that actually converts.

Voice input

Phase 7

Browser MediaRecorder → multipart upload → OpenAI Whisper → transcript fills the chat box. Per-tier monthly transcription quota.

Outcome: hands-free Q&A, accessibility, faster long prompts.

Multi-provider LLM

Phase 8

Provider-agnostic registry (OpenAI / Anthropic / Google), tier-gated picker, graceful degradation when keys are missing, Cursor-style dropdown with locked-state hints.

Outcome: students pick the right brain for the job.

Notifications & reminders

Phase 9

APScheduler cron (Mon + Thu 09:00 UTC), HTML email + in-app bell with unread badge, opt-out preferences, idempotent multi-worker lock.

Outcome: re-engagement and upgrade nudges without spam.

Test & Research modes + clearer trust signals

Phase 10

Two new AI modes (quiz me / research synthesis), revamped confidence heuristic with bail-phrase detection, page-level citations, X icon, refreshed feature bento.

Outcome: turns a chatbot into a study + research toolkit.

Frontend stack

Tool / library	Where it lives	What it does for users
React 18	src/*.jsx	Component model behind every page and widget.
Vite 5	vite.config.js	Instant dev reload, optimised production bundle.
React Router 6	src/App.jsx	Client-side routing — no full page reloads when navigating.
Bootstrap 5 + react-bootstrap	Profile, Auth pages	Form layouts and modals for account flows.
lucide-react	icons everywhere	Consistent line icons; replaced with custom SVG for brand marks.
Custom SVG (SocialIcons)	src/components/SocialIcons.jsx	Stable Instagram / LinkedIn / GitHub / X icons unaffected by lib changes.
@azure/msal-browser	MicrosoftSignInButton.jsx	Microsoft Sign-In popup flow.
Google Identity Services	GoogleSignInButton.jsx	One-tap Google login.
Facebook SDK	FacebookSignInButton.jsx	Facebook OAuth login.
Stripe.js (Checkout redirect)	ProfilePage.jsx	PCI-safe upgrade flow without handling card data ourselves.
MediaRecorder API	ChatPage.jsx	Records mic audio in the browser for voice input.
sessionStorage	useModels.js, theme	Remembers picked model and theme across reloads.
Custom CSS animation	MainPage.css, ChatPage.css	Frosted navbar, conveyor belts, fleeing dots, glow CTA, animated counters.

FRONTEND PAGES

Page	Purpose
MainPage.jsx	Marketing front page — hero, features, footer.
Login.jsx / Signup.jsx	Auth with email/password + 3 OAuth providers, Turnstile captcha.
ForgotPassword.jsx	Email-code password reset.
ChatPage.jsx	Core RAG chat with mode picker, model picker, voice, citations.
UploadPage.jsx	Drag-and-drop document upload.
ProfilePage.jsx	Plan, usage meters, billing portal, upgrade CTA, account.
PaymentReturn.jsx	Stripe success/cancel landing page.
AdminLogin.jsx / AdminDashboard.jsx	Operator console — KPIs, charts, user search, ban.

Backend stack

Tool / library	Version	What it does for users
FastAPI	0.109.0	Async HTTP API serving every chat / upload / auth call.
Uvicorn	0.27.0	ASGI process running FastAPI in the Docker container.
Pydantic + pydantic-settings	2.9 / 2.5	Request/response validation, typed env config.
python-multipart	0.0.6	File and audio upload parsing.
LangChain	0.2.16	Glue between docs, embeddings, vector store, and LLMs.
langchain-openai / -anthropic / -google-genai	0.1.x	Provider clients used by the LLM registry.
FAISS-cpu	1.8.0	Per-user vector index for fast semantic search over uploads.
pypdf, docx2txt, TextLoader	—	PDF / Word / TXT parsing for the upload pipeline.
tiktoken	0.7.0	Counts tokens before each LLM call, prevents context overflow.
numpy	1.26.4	Backbone for FAISS vectors and similarity math.
PyMySQL + cryptography	1.1 / 42.0	MySQL 8 driver with caching_sha2_password support.
bcrypt	4.1.2	Password hashing for email/password accounts.
PyJWT[crypto]	2.9	Verifies Microsoft ID tokens against Microsoft's JWKS.
google-auth	2.36	Verifies Google Sign-In ID tokens.
stripe	9.11	Subscription, customer, webhook, and portal calls.
slowapi	0.1.9	Per-IP rate limits on auth and chat endpoints.
APScheduler	3.10.4	Cron triggers for the bi-weekly reminder fan-out.
httpx	0.27.2	Async HTTP client used by SDKs and OAuth verification.

Tool / library	Version	What it does for users
python-dotenv	1.0.0	Loads backend/.env in dev and inside the container.

BACKEND SERVICES (ONE FILE EACH, SINGLE RESPONSIBILITY)

`chat_service.py`
RAG pipeline — retrieval, prompt build, LLM dispatch, confidence, citations.

`document_service.py`
Upload → chunk → embed → FAISS write.

`chat_history_service.py`
Persists sessions and messages to MySQL.

`user_service.py`
Email/password signup, lookup, profile updates.

`session_service.py`
Bearer token issue / verify / revoke.

`password_reset_service.py`
Reset code email + verification.

`google_oauth_service.py`
Google ID-token verification + auto-signup.

`facebook_oauth_service.py`
Facebook OAuth code exchange + auto-signup.

`microsoft_oauth_service.py`
Microsoft ID-token JWKS verification.

`captcha_service.py`
Cloudflare Turnstile token verification.

`payment_service.py`
Stripe Checkout / Portal / webhooks / tier flips.

`quota_service.py`
Monthly counters: chat, uploads, voice.

`voice_service.py`
Whisper transcription with size + MIME guards.

`email_service.py`
SMTP wrapper used for resets, receipts, reminders.

`notification_service.py`
In-app notifications + preferences CRUD.

`reminder_scheduler.py`
APScheduler cron fan-out (study + upgrade).

`reminder_templates.py`
HTML + plaintext bodies for reminders.

`admin_service.py`
Admin dashboard queries + user management.

`llm/registry.py`
Source of truth for model catalog, tier gating, key checks.

AI / ML stack

Five chat models behind one provider-agnostic interface, plus Whisper for voice and OpenAI embeddings for retrieval.

Model	Provider	Tier	Role	User benefit
GPT-4o	OpenAI	Free	Default chat brain	Fast, capable answers for everyday Q&A.
GPT-5	OpenAI	Advanced	Top-tier reasoning	Hardest research and synthesis questions.
Claude Opus 4.7	Anthropic	Advanced	Long-context / nuance	Careful reasoning over big documents.
Claude Sonnet 4.6	Anthropic	Regular	Balanced workhorse	Quality close to Opus, cheaper response.
Gemini 2.5 Pro	Google	Free	Free-tier alt brain	Strong free option when OpenAI is rate-limited.
Whisper (whisper-1)	OpenAI	Free → Advanced	Speech-to-text	Voice input → text in the chat box.
text-embedding-3-small	OpenAI	All tiers	Doc embeddings	Powers semantic search inside uploads.

How tier gating works

The catalog in `app/services/llm/registry.py` marks every model with a `min_tier` and a provider key. The `/api/models` endpoint returns the full list to every user with `available` + `locked_reason`, so locked models stay visible (greyed out) and act as built-in upsells.

Data layer

MySQL 8.0 (relational state)

FAISS (vector state)

One FAISS index per user under `backend/faiss_store/`, persisted via the `faiss_data` Docker volume so uploads survive container restarts.

Stores 1536-dim embeddings of every uploaded chunk, with the source filename, page number, and original passage attached as metadata so citations point at "page X" instead of "chunk 42".

FILE STORAGE

Raw uploads live alongside the FAISS volume, isolated per user so `document_service.py` can rebuild the index on demand.

Table	What it stores
users	Account, password hash, tier, OAuth provider link.
user_sessions	Bearer tokens, last-seen, expiry.
admin_sessions	Operator login tokens (DB-backed, not in-memory).
password_reset_codes	Time-boxed reset codes.
chat_sessions	Per-user conversation threads.
chat_messages	Every Q/A turn with model, mode, citations.
user_usage_daily	Per-day rollups for the admin charts.
user_quota_usage	Per-month chat / upload / voice counters.
notifications	In-app bell items with kind, title, link, read_at.

Table	What it stores
notification_preferences	User opt-outs for study + upgrade emails.
reminder_runs	Idempotent lock so multi-worker deploys don't double-send.

Pipelines

Each major user action is implemented as a documented pipeline with guard rails.

RAG query pipeline

request → auth check → quota check → mode dispatch → embed query → FAISS top-K → build prompt with grounding → call selected LLM via registry → calculate confidence → format page-level citations → save to `chat_messages` → respond.

Benefit: low-hallucination answers with the page they came from.

Document upload pipeline

upload → MIME + size guard (`MAX_UPLOAD_BYTES`) → loader (pypdf / docx2txt / text) → chunk (`CHUNK_SIZE` / `CHUNK_OVERLAP`) → embed (`text-embedding-3-small`) → write to per-user FAISS index → bump quota counter.

Benefit: students can drop in lecture notes and start asking.

Voice input pipeline

MediaRecorder blob → multipart POST `/api/voice/transcribe` → `MAX_AUDIO_BYTES` guard → MIME whitelist → AsyncOpenAI Whisper → transcript JSON → fills the chat input → quota counter bump.

Benefit: hands-free, accessibility-friendly Q&A.

Auth pipeline

signup form → Turnstile verify → bcrypt hash → users row → bearer token → session row → response. OAuth variant: ID-token verify → upsert user → same session issuance.

Benefit: low-friction signup, abuse protection, single token everywhere.

Payment pipeline

upgrade click → backend creates Stripe Checkout session → redirect → Stripe webhook → update tier on `users` row → reset quotas. Customer Portal link for self-service downgrade / cancel / payment update.

Benefit: PCI-safe upgrades, no card data ever hits our servers.

Reminder pipeline

APScheduler cron (Mon + Thu 09:00 UTC) → `claim_run()` on `reminder_runs` → paginate users → respect `notification_preferences` → SMTP HTML email + insert into `notifications` → `mark_run_finished`.

Benefit: gentle re-engagement + Advanced upsell, never duplicated.

Auth, security & infra

Auth & security

Infrastructure

Component	Role
AWS EC2	Single VM hosting docker compose.
Docker + docker-compose	Reproducible 4-service stack.
MySQL 8 container	Persistent state, healthchecked.
phpMyAdmin container	Browser DB admin (port 8080).
FastAPI backend container	Healthchecked on /health.
nginx + React frontend container	Static SPA + same-origin /api proxy.
faiss_data + mysql_data volumes	Survive rebuilds and container restarts.
env_file + per-service env vars	Secrets out of git, reusable across envs.
Git + GitHub	Source of truth, branch-based deploy.

Layer	User benefit
bcrypt password hashing	Compromised DB ≠ compromised passwords.
Bearer-token sessions in MySQL	Sign out on one device kills only that token.
Google / Facebook / Microsoft OAuth	Skip password hassle, faster signup.
Cloudflare Turnstile CAPTCHA	Blocks bot signups without captcha pain.
slowapi rate limiting	Per-IP throttling on auth + chat = spam shield.
MAX_UPLOAD_BYTES + MAX_AUDIO_BYTES	Stops giant uploads from degrading service.
IDOR-hardened admin queries	Operators can't read another user's data accidentally.
AdGuard-proof admin CSS classnames (.adm-*)	Dashboard renders even with ad blockers.
Strong password policy + phone validation	Common-password rejection at signup.
Per-user FAISS isolation	One student never sees

Layer	User benefit
	another's documents.

User-facing features

Feature	Tier	User benefit
Document Q&A with citations	All	Get the answer + the page it came from.
Deterministic AI mode	All	Strict, no-hallucination answers from your docs.
Exploratory AI mode	All	Grounded answer + broader context, even with no uploads.
Test mode (quizzes from docs)	All	Self-quiz with MCQ / short-answer / true-false.
Research mode (academic synthesis)	All	Structured notes and inline citations for university work.
Multi-model picker	All (gated)	Pick GPT-4o, Gemini, Sonnet, Opus, GPT-5 by tier.
Voice input	All (quota)	Speak instead of type — Whisper transcription.
Confidence chip with tooltip	All	See how sure the model is, and why.
Page-level citations (filename — page X)	All	Trust + verifiability; no scary chunk indices.
Per-user document isolation	All	Privacy by default.
Stripe-managed subscriptions	Regular / Advanced	Self-service upgrade, downgrade, cancel.
Profile usage meters	All	Live remaining quota for chat / uploads / voice.
Notifications bell + dropdown	All	Reminders and account events without leaving the app.
Bi-weekly study reminders (email + in-app)	Free	Gentle nudge to keep using the platform.
Bi-weekly upgrade reminders (email + in-app)	Free	Surface the Advanced plan twice a week.
Notification preferences page	All	Opt out per channel + per kind.
Password reset by email code	All	Self-service recovery.
3 OAuth providers	All	One-click signup with Google / Facebook / Microsoft.

Feature	Tier	User benefit
Light + dark theme toggle	All	Comfortable reading day or night.
Sticky frosted navbar + animated hero	Visitors	Front page that converts.
Admin dashboard (KPIs, charts, search, ban)	Operators	Run the platform without SSHing into MySQL.

API surface (high level)

Endpoint	Purpose
GET /health	Container healthcheck for Docker.
POST /api/auth/signup, /login, /oauth/*	Account creation + sign-in (email + 3 OAuth).
POST /api/auth/forgot-password, /reset-password	Email-code password recovery.
GET /api/auth/me	Current user + tier.
POST /api/documents/upload	Upload PDF / DOCX / TXT into FAISS.
GET /api/documents, /api/documents/count	List + count user's docs.
DELETE /api/documents/{id}	Remove a document and its vectors.
POST /api/chat/query	RAG turn — accepts mode + model fields.
GET /api/chat/sessions, /api/chat/sessions/{id}	Chat history.
GET /api/models	Tier-aware model catalog for the picker.
POST /api/voice/transcribe	Whisper speech-to-text endpoint.
GET /api/usage/quota	Per-month chat / upload / voice counters.
GET /api/notifications, /unread-count, /preferences	Bell dropdown data + opt-out.
POST /api/notifications/{id}/read, /read-all	Mark items read.
POST /api/payments/checkout, /portal	Stripe Checkout + Customer Portal redirects.
POST /api/payments/webhook	Stripe webhook → tier flip.
GET /api/admin/*	Admin KPIs, charts, user search, ban.

Bottom line

From a single-model RAG demo, Laboracle has grown into a 4-service Docker deployment with 5 LLMs behind one registry, voice input, four AI modes, three OAuth providers, Stripe subscriptions, an admin console, and a notification system. Every layer was added behind a feature flag or graceful-degradation guard, so the app keeps running even when individual provider keys are missing.